

Week 5 — Isostasy Problem: Oceanic vs Continental Crust

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Part A: Crust-only Airy Isostasy

Given:

oceanic crust thickness: $h_{oc} = 7 \text{ km}$

ocean depth: $h_w = 3.7 \text{ km}$

seawater density: $\rho_w = 1030 \text{ kg m}^{-3}$

continental crust thickness: $h_{cc} = 40 \text{ km}$

continental crust density: $\rho_{cc} = 2850 \text{ kg m}^{-3}$

continental elevation: $\varepsilon = 0.8 \text{ km}$

mantle density: $\rho_a = 3330 \text{ kg m}^{-3}$

Instructions: Assume isostatic compensation occurs solely through crustal thickness. Compute the density of the oceanic crust, ρ_{oc} , that balances the oceanic and continental columns. Compare your result with typical oceanic crust densities ($\sim 2900\text{--}3000 \text{ kg m}^{-3}$) and discuss why the crust-only assumption may fail.

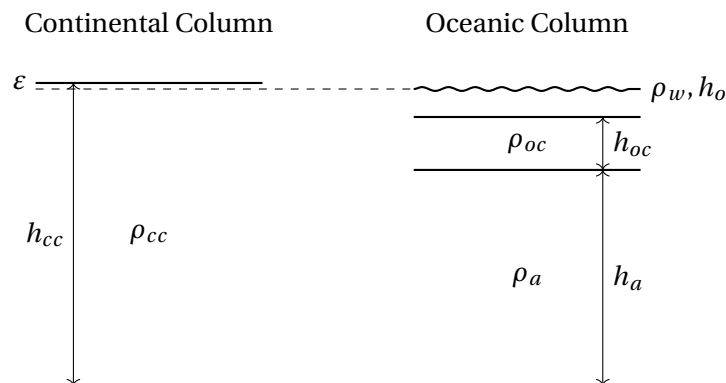


Figure 1: Comparison of isostatic columns for the continental and oceanic crust for Part A.

Questions:

1. Compute the required oceanic crust density ρ_{oc} .
2. Compare with expected oceanic crust densities and discuss why the simple crust-only model may not yield realistic values.

Part B: Two-layer Lithosphere Model

Given additional information:

Oceanic lithospheric mantle thickness $h_{om} = 75 \text{ km}$

Oceanic lithospheric mantle density: $\rho_{om} = 3330 \text{ kg m}^{-3}$

Continental lithospheric mantle thickness: $h_{cm} = 150 \text{ km}$

Continental lithospheric mantle density: $\rho_{cm} = ?$

Asthenosphere density: $\rho_a = 3320 \text{ kg m}^{-3}$

Instructions: Let's add a two-layered lithosphere using average lithospheric thicknesses for the oceanic and continental settings. Our goal will be to see what we can learn about the relative density difference between continental and oceanic mantle lithosphere. In the process, we'll show how including the mantle lithosphere resolves the unrealistic crust-only result.

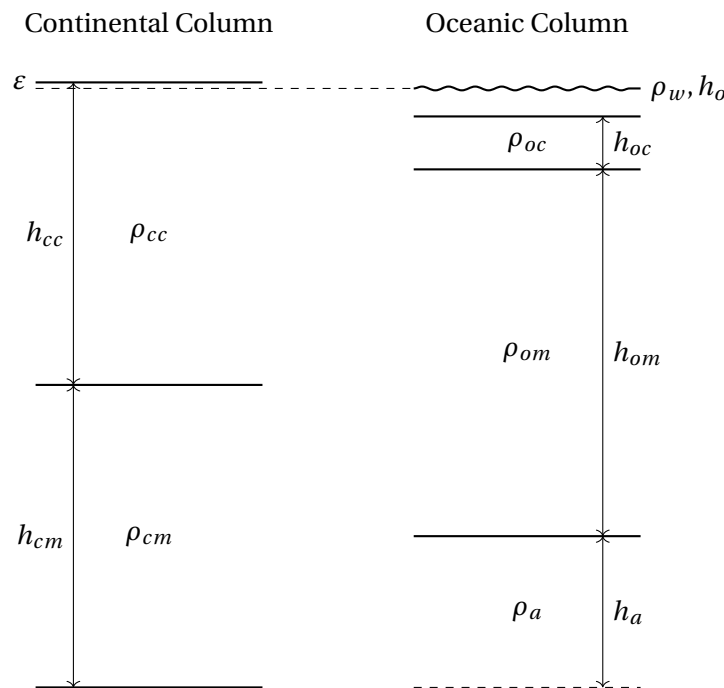


Figure 2: Comparison of isostatic columns for the continental and oceanic crust for Part B.

Table 1: Modal proportions of minerals in the oceanic and continental lithospheres.

Mineral	Oceanic	Continental
olivine	75	60
orthopyroxene	25	17
clinopyroxene	10	11
garnet	0	12

Oceanic lithospheric mantle is assumed to be composed of depleted peridotite with a density of $\rho_{om} = 3330 \text{ kg m}^{-3}$. Continental lithospheric mantle may contain a higher proportion of garnet.

Density of depleted mantle (peridotite): $\rho_d = 3330 \text{ kg m}^{-3}$

Density of garnet: $\rho_g = 3600 \text{ kg m}^{-3}$

Questions:

1. Setup the isostatic equations for a two-layered mantle lithosphere. If we assume an average oceanic crustal density $\rho_{oc} = 2900\text{--}3000 \text{ kg m}^{-3}$, what density of the continental mantle lithosphere?
2. Explain how including the mantle lithosphere resolves the unrealistic density obtained in the crust-only case.
3. Discuss the role of garnet fraction in the continental lithospheric mantle for the density contrast. Compute the percentage difference garnet between the oceanic and continental lithosphere and compare your result with the modal proportions in Table 1.